



# SMART TRAFFIC MANAGEMENT SYSTEM

P R E S E N T A T I O N 2 0 2 6



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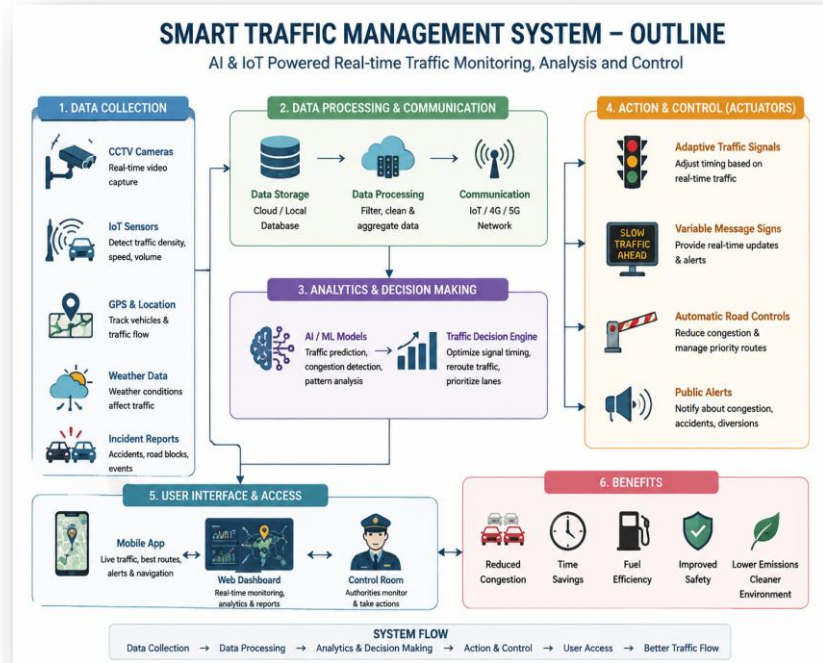
Guide

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**EDUNET FOUNDATION**



# OUTLINE

- Abstract of the Project
- Problem Statement
- Proposed Solution
- System Architecture
- Live Demo of the Project
- Embedded Video of Project
- Conclusion
- Future Scope



## Abstract

The Smart Traffic Management System is an intelligent and data-driven solution developed to monitor, analyze, and manage traffic conditions efficiently in urban areas. The system uses traffic-related data such as vehicle count, vehicle type, vehicle speed, time of day, weather conditions, road conditions, and congestion levels to understand traffic patterns. The collected data is preprocessed and analyzed using data analytics and machine learning techniques to identify existing traffic conditions and predict future congestion levels. Based on the analysis, the system can support better traffic signal management, route planning, and traffic control decisions. The main goal of the proposed system is to reduce traffic congestion, vehicle waiting time, travel time, fuel consumption, and harmful emissions. It also helps improve road safety and provides useful insights for transportation authorities to make effective and timely decisions. By combining real-time traffic monitoring, data analysis, and machine learning, the Smart Traffic Management System provides an efficient and scalable approach for improving traffic flow and creating a safer, faster, and more sustainable transportation environment.

# Problem Statement

## Problem Statement – Smart Traffic Management System

Traffic congestion is one of the major problems in urban areas due to the increasing number of vehicles, limited road capacity, poor traffic signal management, and unpredictable traffic patterns. Traditional traffic management systems often depend on fixed signal timings and manual monitoring, which may not respond effectively to changing traffic conditions. This results in increased vehicle waiting time, longer travel time, fuel consumption, air pollution, and the possibility of accidents. Therefore, there is a need for an intelligent system that can analyze traffic-related data and identify or predict congestion levels. The proposed **Smart Traffic Management System** uses data analytics and machine learning techniques to analyze factors such as vehicle count, vehicle type, speed, time of day, weather, and road conditions. The system aims to provide accurate traffic congestion predictions and useful insights that can help authorities optimize traffic flow, improve road safety, reduce delays, and make traffic management more efficient.

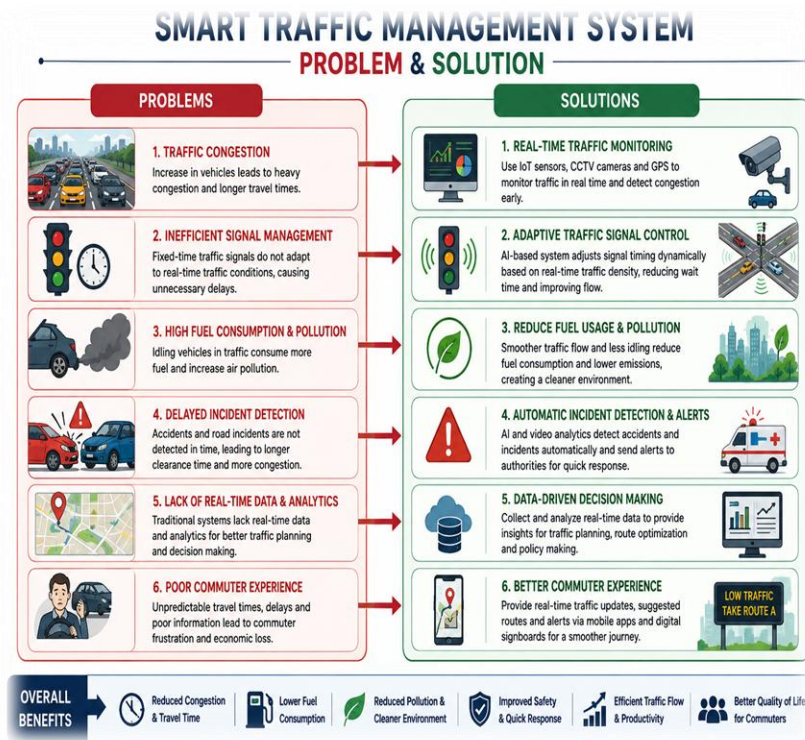




# Proposed Solution

## Proposed Solution – Smart Traffic Management System

The proposed Smart Traffic Management System uses **data analytics and machine learning** to monitor, analyze, and predict traffic conditions. Traffic data such as vehicle count, vehicle type, speed, time of day, weather conditions, road conditions, and previous congestion levels are collected and processed. The data is cleaned, transformed, and analyzed to identify important traffic patterns. Machine learning algorithms are then applied to **predict congestion levels** such as low, medium, and high. Based on the predicted traffic condition, the system can support **dynamic traffic signal control, better route planning, and efficient traffic flow management**. The system provides useful insights through visualizations and dashboards, helping traffic authorities make faster and more informed decisions. The proposed solution aims to reduce **traffic congestion, waiting time, travel time, fuel consumption, and pollution**, while improving road safety and overall transportation efficiency.



# System Architecture

## DATA COLLECTION 1

The system collects traffic data such as vehicle count, vehicle type, speed, time of day, weather conditions, road conditions, and previous congestion levels. This data is used to understand traffic patterns.

## MACHINE LEARNING PREDICTION 3

Machine learning algorithms are applied to analyze traffic patterns and predict congestion levels such as Low, Medium, and High. The prediction helps identify possible traffic problems in advance.

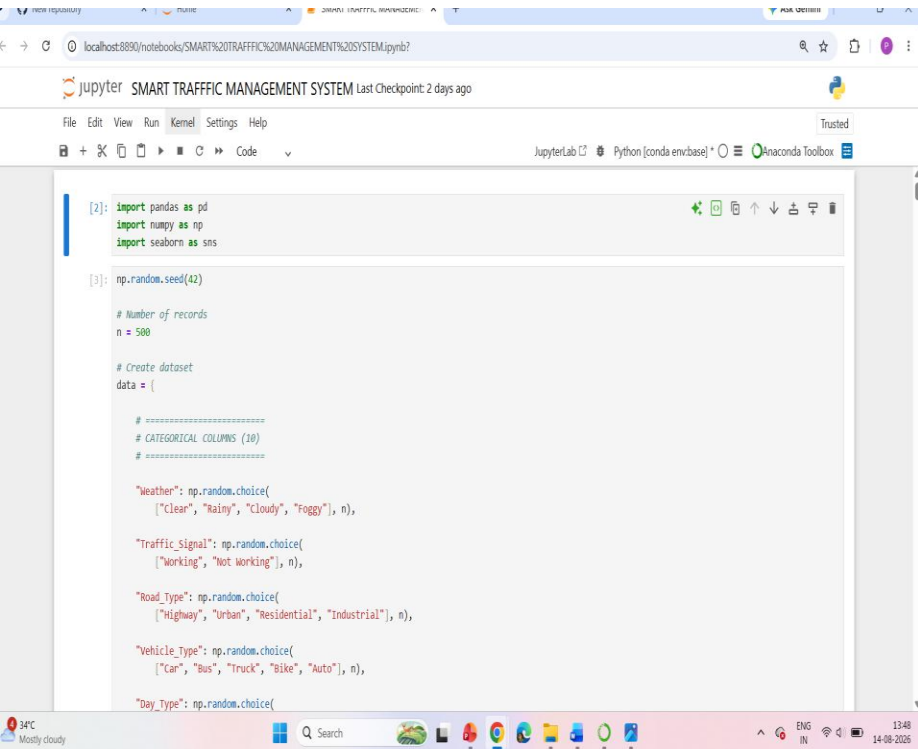
## DATA PROCESSING 2

The collected data is cleaned and prepared by handling missing values, removing duplicate records, and converting categorical data into suitable formats. This improves the quality and accuracy of the analysis.

## TRAFFIC MANAGEMENT & VISUALIZATION 4

The predicted results are displayed using charts, graphs, and dashboards. These insights help traffic authorities optimize traffic signals, plan routes, reduce waiting time, and improve overall traffic flow.

# Images of Project Demo



Jupyter SMART TRAFFIC MANAGEMENT SYSTEM Last Checkpoint: 2 days ago

```
[2]: import pandas as pd
import numpy as np
import seaborn as sns

[3]: np.random.seed(42)

# Number of records
n = 500

# Create dataset
data = {

# =====
# CATEGORICAL COLUMNS (10)
# =====

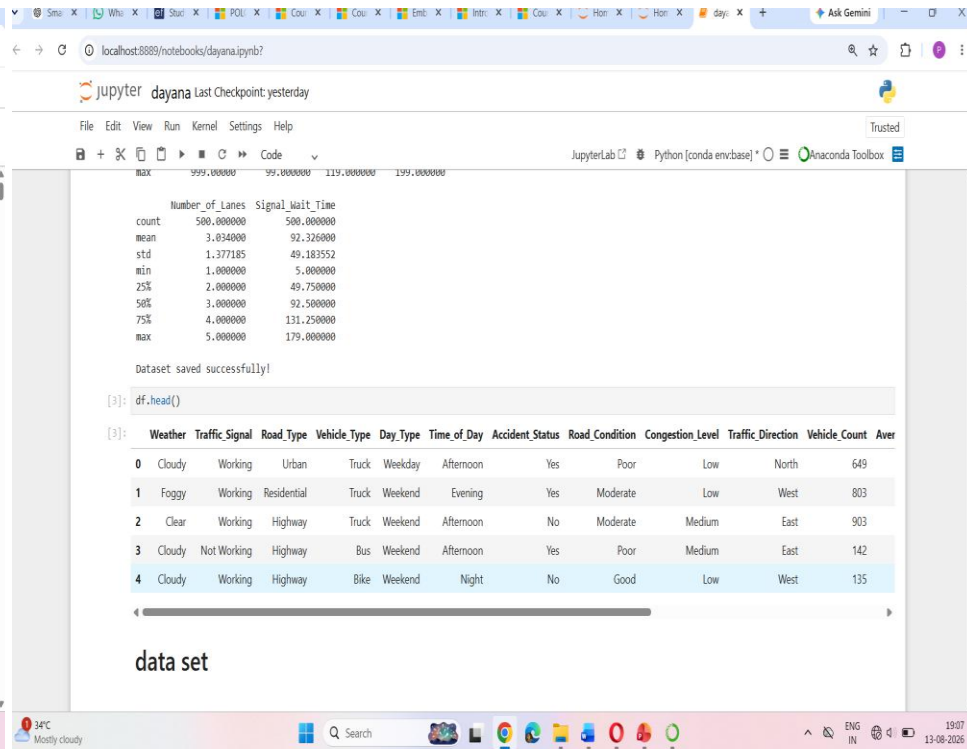
"weather": np.random.choice(
    ["Clear", "Rainy", "Cloudy", "Foggy"], n),

"Traffic_Signal": np.random.choice(
    ["Working", "Not Working"], n),

"Road_Type": np.random.choice(
    ["Highway", "Urban", "Residential", "Industrial"], n),

"Vehicle_Type": np.random.choice(
    ["car", "Bus", "Truck", "Bike", "Auto"], n),

"Day_Type": np.random.choice(
```



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```
MAX 399.000000 99.000000 119.000000 199.000000
```

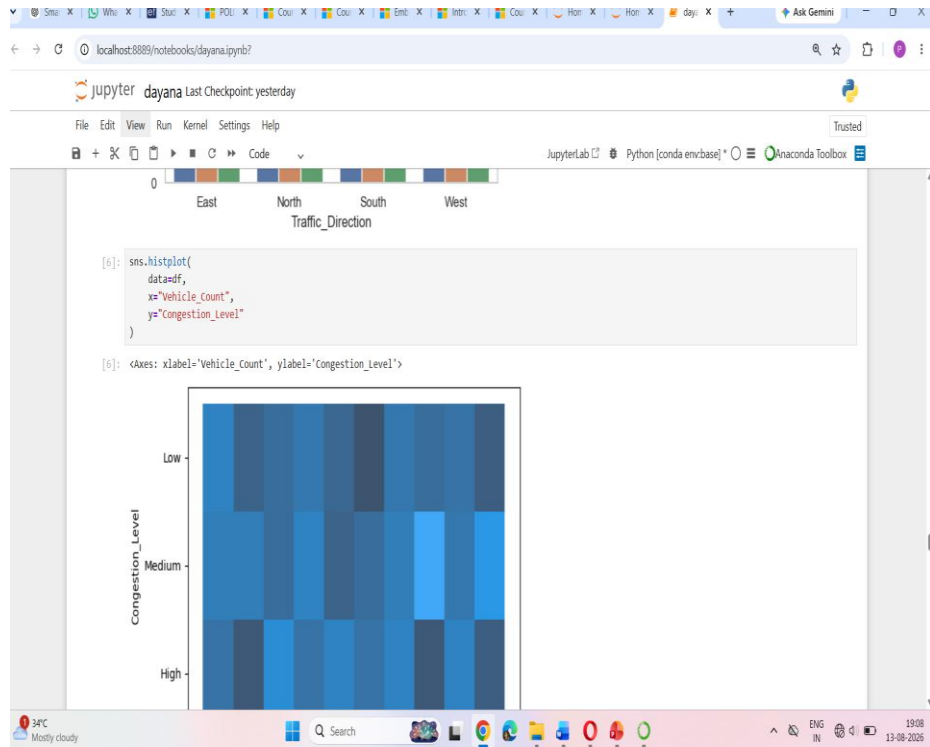
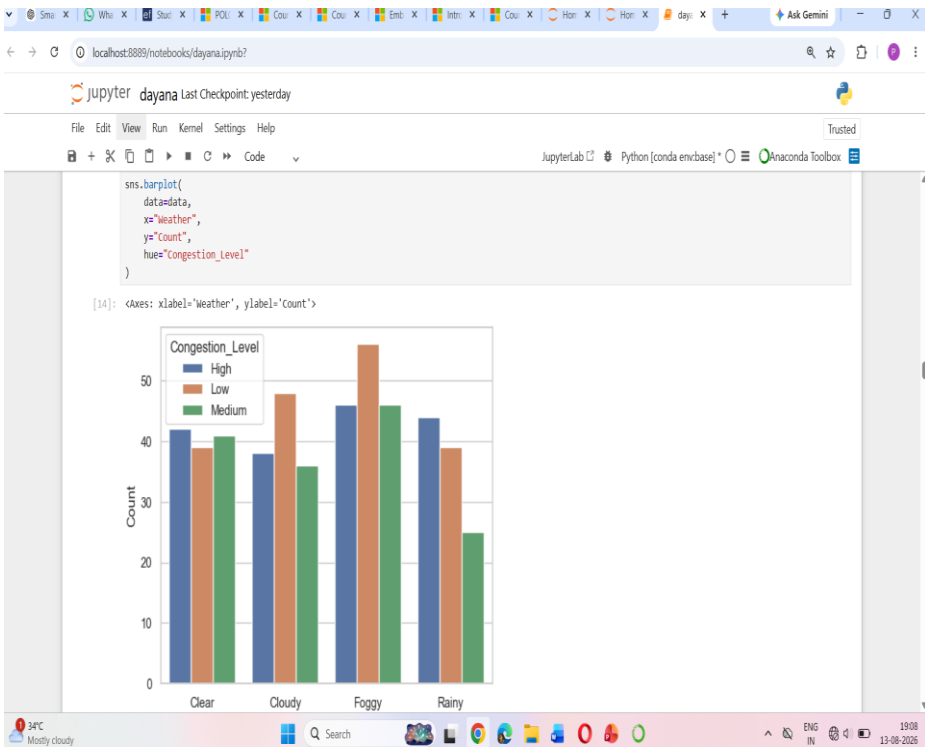
|       | Number_of_Lanes | Signal_Wait_Time |
|-------|-----------------|------------------|
| count | 500.000000      | 500.000000       |
| mean  | 3.034000        | 92.326000        |
| std   | 1.377185        | 49.183552        |
| min   | 1.000000        | 5.000000         |
| 25%   | 2.000000        | 49.750000        |
| 50%   | 3.000000        | 92.500000        |
| 75%   | 4.000000        | 131.250000       |
| max   | 5.000000        | 179.000000       |

Dataset saved successfully!

```
[3]: df.head()
```

|   | Weather | Traffic_Signal | Road_Type   | Vehicle_Type | Day_Type | Time_of_Day | Accident_Status | Road_Condition | Congestion_Level | Traffic_Direction | Vehicle_Count | Aver |
|---|---------|----------------|-------------|--------------|----------|-------------|-----------------|----------------|------------------|-------------------|---------------|------|
| 0 | Cloudy  | Working        | Urban       | Truck        | Weekday  | Afternoon   | Yes             | Poor           | Low              | North             | 649           |      |
| 1 | Foggy   | Working        | Residential | Truck        | Weekend  | Evening     | Yes             | Moderate       | Low              | West              | 803           |      |
| 2 | Clear   | Working        | Highway     | Truck        | Weekend  | Afternoon   | No              | Moderate       | Medium           | East              | 903           |      |
| 3 | Cloudy  | Not Working    | Highway     | Bus          | Weekend  | Afternoon   | Yes             | Poor           | Medium           | East              | 142           |      |
| 4 | Cloudy  | Working        | Highway     | Bike         | Weekend  | Night       | No              | Good           | Low              | West              | 135           |      |

data set





## Conclusion

The Smart Traffic Management System provides an efficient and intelligent approach to managing increasing traffic congestion. By using data analytics and machine learning, the system can analyze important traffic factors such as vehicle count, vehicle type, speed, time of day, weather, and road conditions to predict congestion levels. The system helps reduce traffic delays, waiting time, fuel consumption, and vehicle emissions while improving road safety and traffic flow. The proposed system can support traffic authorities in making faster and data-driven decisions. Overall, it provides a smart, reliable, and scalable solution for improving urban transportation and creating a more efficient and sustainable traffic management system.



# Future Scope

**Real-time traffic monitoring:** Integrate IoT sensors, CCTV cameras, and GPS data to monitor traffic conditions in real time.

**AI-based traffic prediction:** Use advanced machine learning and deep learning models to predict congestion more accurately.

**Smart traffic signals:** Develop adaptive traffic signals that automatically change timing based on real-time traffic density.

**Emergency vehicle priority:** Give priority to ambulances, fire engines, and other emergency vehicles to reduce response time.



# Thank You